

RACHEL ATKEISON

Computer Science | Software Engineering | Real-Time Systems, Frontend, Audio & Creative Technology

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Python | C++ | JavaScript | TypeScript | React | C# | Java | SQL | Unity | JUCE | PySide6 | WebSockets | Git/GitHub

SUMMARY

I am a software engineering candidate with a rare ability to connect systems thinking, creative technology, audio engineering, visual design, and clear communication. My work is not limited to one lane: I have built real-time collaboration tools, C++ audio software, interactive 3D environments, visualization engines, and polished frontend experiences, each shaped by a desire to make complex technology feel understandable and alive. With a Computer Science foundation, 12 years of art/design experience, music technology projects, and experience tutoring both programming and writing, I bring the perspective of an engineer who can also think deeply about users, presentation, and meaning. I am seeking a software engineering opportunity where I can build useful products, grow quickly, and contribute the kind of technical and creative perspective that makes teams stronger.

EDUCATION

Kennesaw State University — Kennesaw, GA

B.S. Computer Science | Dean's List Recipient | Expected 2027

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Object-Oriented Programming, Database Systems, Network Communications, User-Centered Design, Computer Graphics

TECHNICAL SKILLS

Languages: Python, C++, C#, JavaScript, TypeScript, Java, SQL, HTML/CSS

Frameworks & Tools: React, PySide6, Qt, JUCE, Unity, Java Swing, WebSockets, asyncio, NumPy, PyQtGraph

Technical Areas: Real-time systems, frontend development, UI/UX, distributed state, object-oriented programming, audio/DSP, MIDI, data visualization, cloud-connected interfaces

Development: Git/GitHub, CMake, AWS Lambda, AWS IoT Core, S3, Figma, VS Code, debugging, documentation, product thinking

INTERACTIVE MEDIA & CREATIVE ENGINEERING PROJECTS

Aureine Live | Real-Time Distributed Music Collaboration Engine

Python, PySide6, asyncio, WebSockets, NumPy, sounddevice

- Built a networked music collaboration system with shared timing, session state, user activity, and visual feedback.
- Modularized networking, synchronization, jitter buffering, audio synthesis, UI rendering, and replay tools.

LumenBloom | Polyphonic Synthesizer Plugin

C++, JUCE, CMake, DSP, VST3, MIDI

- Built a C++/JUCE synth plugin with polyphony, oscillators, envelopes, filters, modulation, effects, presets, and MIDI input.
- Structured real-time audio logic across voices, signal flow, DSP modules, UI state, and DAW-ready behavior.

CloudStage MR Studio | Interactive Spatial Audio Environment

Unity, C#, JUCE, C++, MIDI, Spatial Audio

- Built a 3D performance environment with playable instruments, spatial audio, reactive lighting, and role-based interaction.
- Connected Unity controls to an external JUCE synthesis bridge for real-time audio generation.

Y-Control Hub | Cloud-Connected Dashboard Prototype

React, JavaScript, AWS Lambda, AWS IoT Core, S3

- Built a React dashboard prototype for viewing and controlling connected device settings.
- Connected frontend controls to cloud-service concepts, device state, event updates, and dashboard interactions.

Pizza Time Pizza | Customer Ordering Application

Java, Java Swing, Object-Oriented Programming, UI Flow

- Built a desktop ordering app with login, menu browsing, cart management, item selection, and checkout.
- Practiced requirements planning, class design, object-oriented structure, and customer-facing UI flow.

EXPERIENCE

Aureine Audio Systems | Independent Software Developer

Independent Software Developer | 2025–Present

- Build self-directed software projects across real-time collaboration, C++ audio tools, interactive media, visualization systems, cloud dashboards, and frontend interfaces.
- Take projects from concept to working demo by planning user flows, choosing architecture, writing code, testing behavior, documenting features, and improving presentation.
- Combine software engineering, design, music, and visual systems to create projects that are functional, understandable, and easy to evaluate.

Computer Science & English Tutor | Kennesaw State University

Kennesaw State University | Dec. 2021 – May 2025

- Tutored students in programming fundamentals, debugging, object-oriented thinking, academic writing, and problem-solving.
- Helped students break confusing assignments into smaller steps and adapted explanations for different learning styles and skill levels.
- Built strong communication and mentorship skills by explaining technical concepts with patience, clarity, and practical examples.

SIGNATURE STRENGTH

My strength is turning complexity into something visible, usable, and memorable. I build like an engineer, shape like a designer, explain like a tutor, and keep refining until the work feels clear enough for someone else to believe in.